

EXPERMENT 2

2. AIM: Study of Network devices in detail and connect the computers in Local Area Network.

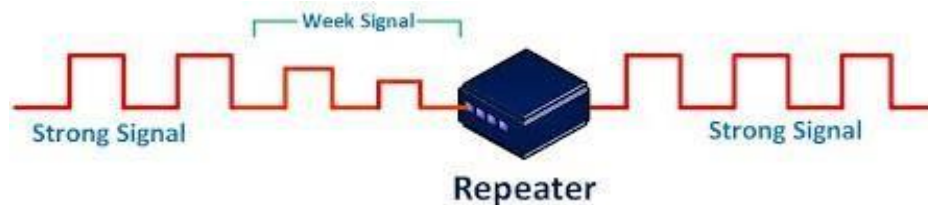
Study of Network Devices in Detail and connect the computers in Local Area Network

a) Study of Network Devices in Detail

1. Repeater
2. Hub
3. Switch
4. Bridge
5. Router
6. GateWay

1. Repeater:

A Repeater is a network device that operates at the Physical Layer (Layer 1) of the OSI model. It receives a weak signal, regenerates or strengthens it, and retransmits it as a strong signal. This helps extend the communication distance and prevents signal loss over long transmission paths.



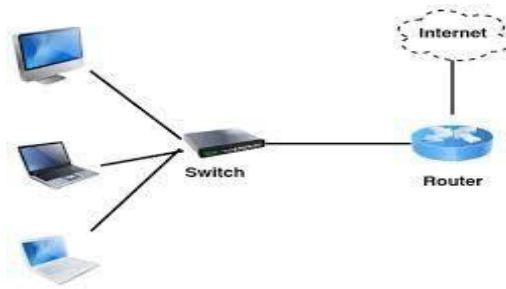
2. Hub: A Hub is a network device used to connect multiple computers or devices in a network. It works at the Physical Layer (Layer 1) of the OSI model. When a hub receives data from one computer, it broadcasts the data to all connected computers. Therefore, all devices share the same network segment.



3. Switch:

A Switch is a networking device that connects multiple computers and devices in a Local Area Network (LAN). It operates at the Data Link Layer (Layer 2) of the OSI model.

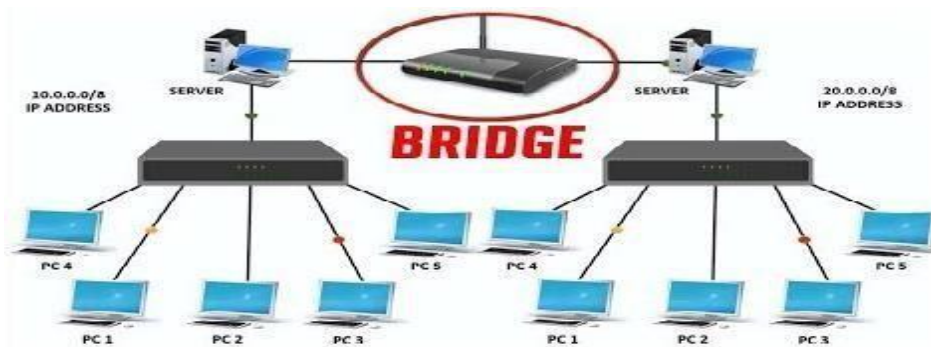
Unlike a hub, a switch sends data only to the intended destination device by using MAC addresses, which makes communication faster and more efficient.



4. Bridge:

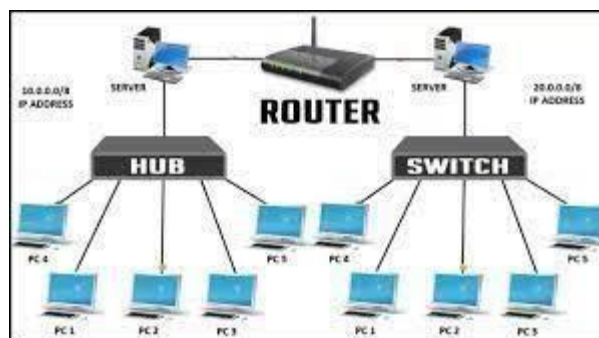
A Bridge is a network device that connects two or more network segments. It operates at the Data Link Layer (Layer 2) of the OSI model.

A bridge uses MAC addresses to filter and forward data between network segments. It checks incoming data frames and forwards them only when necessary, which helps reduce network traffic.



5. Router:

A Router is a networking device that connects two or more different networks and forwards data packets between them. It operates at the Network Layer (Layer 3) of the OSI model.



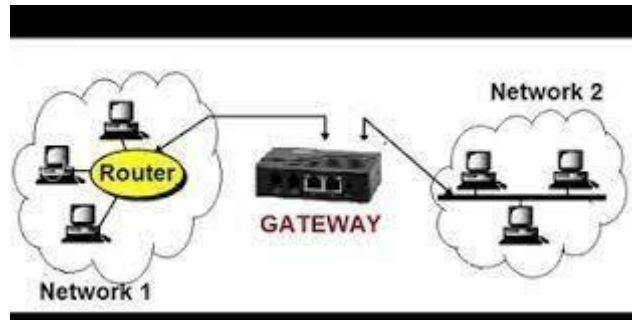
A router uses IP addresses to determine the best path for data to travel from the source network to the destination network. It allows communication between different networks and provides access to the Internet.

6. Gate Way:

A Gateway is a network device that connects two different networks and enables communication between them. It acts as a protocol converter and allows networks using

different communication protocols to exchange data.

A gateway operates at multiple layers of the OSI model, depending on the type of protocol conversion required.



b) Connect the Computer to a Local Area Network (LAN) Using a Mobile Hotspot

Procedure

Step 1: Enable Mobile Hotspot

1. Enable **Personal Hotspot (Mobile Hotspot)**.

Step 2: Connect the Computer

1. Turn on **Wi-Fi** on the computer.
2. Select the mobile hotspot from the list of available Wi-Fi networks.
3. Enter the hotspot password.
4. Click **Connect**.

Step 3: Verify the IP Address

1. Open **Command Prompt**.
2. Type: **ipconfig**
3. Note the following:
 - **IPv4 Address** (Computer IP Address)
 - **Default Gateway** (Mobile Hotspot IP Address)

Example:

IPv4 Address : 192.168.0.107

Subnet Mask : 255.255.255.0

Default Gateway : 192.168.0.1

Step 4: Test the Connection

Type the following command:

ping 192.168.0.1

(Replace **192.168.0.1** with your **Default Gateway**.)

Example Output:

Reply from 192.168.0.1: bytes=32 time=1ms TTL=64

Reply from 192.168.0.1: bytes=32 time=2ms TTL=64

Reply from 192.168.0.1: bytes=32 time=1ms TTL=64

Reply from 192.168.0.1: bytes=32 time=2ms TTL=64

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Observation

Parameter	Value
Computer IP Address	192.168.0.107
Mobile Hotspot IP Address	192.168.0.1

Parameter	Value
Connection Type	Wireless LAN (Wi-Fi)
Ping Result	Successful

Output:

The computer was successfully connected to the Wireless Local Area Network (WLAN) using a mobile hotspot. The network connectivity was verified using the ping command.